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Sample Size Calculation

Purpose

Sample size calculation is performed before data collection to determine the minimum number of participants required for reliable and statistically valid results.

A sample that is too small may fail to detect real associations, while an unnecessarily large sample may waste time and resources.

Common Tools

- OpenEpi (Most commonly used in student research)
 - Raosoft Sample Size Calculator
 - Epi Info
 - G*Power (advanced analyses)
-

Parameters Required

Population Size (N)

The total number of individuals in the target population.

Examples:

- All medical students at Yarmouk University.
- All diabetic patients attending a specific clinic.

If the population size is known, it should be entered into the calculator.

Confidence Level

Represents how confident we are that the sample reflects the population.

Most commonly used: 95%

A 95% confidence level means that if the study were repeated many times, the true population value would fall within the confidence interval in approximately 95% of studies.

Margin of Error (Precision)

Represents the acceptable difference between the sample estimate and the true population value.

Most commonly used: 5%

Smaller margins of error require larger sample sizes.

Expected Prevalence (Expected Frequency)

The estimated percentage of participants expected to have the characteristic being studied.

Example:

If previous studies suggest that 30% of students experience burnout, then the expected prevalence is 30%.

If no reliable estimate is available, use **50%**.

Why Use 50%?

A prevalence of 50% produces the largest required sample size and is therefore considered the safest and most conservative choice.

Finite Population Correction (FPC)

What Is It?

Finite Population Correction is used when the total population size is known and limited.

Instead of assuming an infinite population, the calculation adjusts for the actual population size.

Why Is It Important?

For small populations, fewer participants are needed than the standard sample size of 384.

Examples

Using:

- Confidence Level = 95%
- Margin of Error = 5%
- Expected Prevalence = 50%

Population Size	Required Sample Size
500	≈ 218
1000	≈ 278
2000	≈ 323
5000	≈ 357

Population Size	Required Sample Size
10000	≈ 370
Very Large Population	≈ 384

As the population becomes larger, the sample size gradually approaches 384.

Example Calculation

Suppose:

- Population size = 2,000 students
- Confidence level = 95%
- Margin of error = 5%
- Expected prevalence = 50%

Using OpenEpi:

Required sample size ≈ 323 participants.

To account for incomplete responses, researchers often increase the target sample by 10–20%.

For example:

$323 \times 1.20 = 388$ participants.

Thus, a target sample of approximately 390 participants may be used.

Reporting Example

The minimum required sample size was calculated using OpenEpi with finite population correction, assuming a population size of 2,000 students, a 95% confidence level, a 5% margin of error, and an expected prevalence of 50%. The calculated minimum sample size was 323 participants.

Important Notes

If Population Size Is Unknown

Use a large-population calculation.

With:

- 95% confidence level
- 5% margin of error
- 50% prevalence

The required sample size is approximately 384 participants.

If Previous Literature Exists

Use the prevalence reported by previous studies rather than automatically choosing 50%.

Increasing the Sample Size

Researchers commonly add 10–20% to compensate for:

- Non-response
 - Missing data
 - Incomplete questionnaires
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Golden Rule

If no previous prevalence estimate is available:

✓ Confidence Level = 95%

✓ Margin of Error = 5%

✓ Expected Prevalence = 50%

✓ Use Finite Population Correction when the population size is known.

This approach is widely accepted and appropriate for most cross-sectional medical research studies.